

Estimation of DC Machine Parameters and Simulations

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I Introduction

The purpose of this report is to analyze the operation and functionality of DC machines, focusing on parameter estimation based on the nominal characteristics of a DC machine and laboratory measurements taken from the armature and excitation circuits. The DC machine under investigation has a power rating of 3 kW and operates at a nominal speed of 1500 rpm. This study aims to provide an accurate simulation model by estimating key machine parameters, including resistance and inductance values, based on experimental data. The primary objective is to process and analyze measurement data to estimate the machine's parameters and then use these parameters to create a detailed Simulink model. Through simulations, this model will be used to verify the steady-state torque-speed characteristics of the machine, ensuring alignment with theoretical expectations. The main section of this project is divided in three categories.

II DC machine Parameters Estimation

This section provides a detailed methodology for processing the real measurement data obtained from laboratory tests. Moreover, some estimated parameters that arise from the nominal characteristics of the machine are going to be used. This section has a paramount importance as the obtained numerical results are going to be used in the model.

The circuit that is presented below was used to obtain the set of measurements. It is the equivalent circuit that presents the armature and the excitation circuit. It should be stated for the measurements that when the armature circuit was energized, the excitation circuit had zero current, so no $Kf\Phi$ was generated and as a result no torque. Also, when the excitation circuit was energized, the armature circuit had zero current, so no torque and no armature current was generated. Another im-

portant thing that I would like to mention for a better understanding of the behavior of the raw data, is the sudden voltage drop across the diode when the switch is opened at the beginning of the discharge phase. This sharp drop is due to the voltage drop inherent in the diode, as in the measurements, there is a real diode that is not treated as an ideal component.

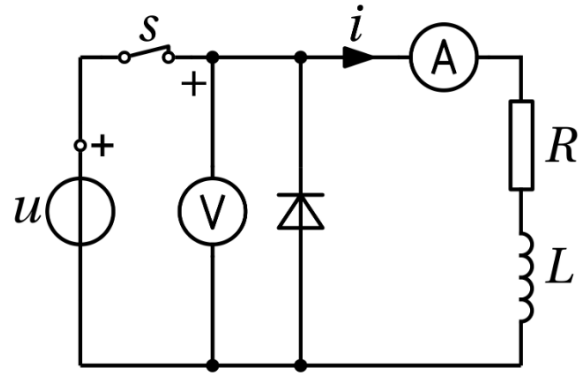


Figure 1: Equivalent circuit of armature and excitation circuit

In each set of measurements, I had different stabilized starting points for both current and voltage. When the switch was turned off, a discharge occurred from the starting point down to 0 A. For the armature current, I collected measurements at 1 A, 2 A, 4 A, 6 A, 8 A, and 10 A, while for the excitation current, the measurements were taken at 0.2 A, 0.4 A, 0.6 A, 0.8 A, and 1 A. As the raw data had a lot of noise a key step was to include the off-set correction in each circuit to estimate the parameters ensuring better accuracy. To correct these offsets after plotting and investigated the raw data, I determined the offsets by analyzing the last 15 percent of each measurement, where both current and voltage were expected to be zero. The calculated offsets were subtracted from the corresponding current and voltage measurements for each data in a try to eliminate this bias. Moreover, the examination of the time series data for each current

level was another one significant step to identify the points at which the discharge phenomenon started (when the switch was turned off). So this time segmentation, divided the time in two parts:

The steady-state section: The current and voltage values are stabilized and remain relatively constant.

The transient-state section: The values fluctuated as the system discharged.

A Nominal values

An analytical description of the motor's nominal parameters is essential. This will extract reference values of the system and necessary information for the simulation which can help to verify certain results obtained from the raw data measurements.

Nominal Parameter	Value	Unit
Voltage of the Armature	220	V
Current of the Armature	17	A
Voltage of the Excitation	145	V
Current of the Excitation	1	A
Power	3	kW
Speed	1500	rpm

For the nominal torque of the machine we have that:

$$T = \frac{P}{\omega} = \frac{3000 \text{ W}}{1500 \text{ RPM}} = \frac{3000}{1500 \times \frac{2\pi}{60}}$$

$$= \frac{3000}{157.0796} = 19.0986 \text{ Nm}$$

For the $K_f \Phi$ we have that:

$$K_f \Phi = \frac{T}{I_a}$$

$$K_f \Phi = 1.1234$$

B Armature Parameters Estimation

Armature Resistance (R_a)

The Armature resistance was estimated by calculating the ratio of the mean voltage to the mean current during the steady-state period. This provided an estimate of the armature resistance for each current level. After that, to increase reliability, I calculated the average resistance across all current levels, which provided a more reliable estimate of the armature resistance.

$$R_a = \frac{\bar{V}_{\text{mean, steady state}}}{\bar{I}_{\text{mean, steady state}}} = 1.5456 \text{ Ohms}$$

Armature Inductance (L_a)

For the estimation of the armature inductance L_a , I followed the provided tips. First of all, I estimated the flux linkage (λ) using the cumulative sum ('cumsum') integration. The integrand represents the voltage drop across the inductor: $V - R \cdot I$. Once the flux linkage values were obtained for different armature currents, a linear regression was performed to establish the relationship between the flux linkage and current: $\lambda = L \cdot I$. By fitting a straight line to this data using a first-degree polynomial, I determined the inductance (L_a) of the armature circuit by the slope of the line.

Thus,

$$L_a = 16 \text{ mH}$$

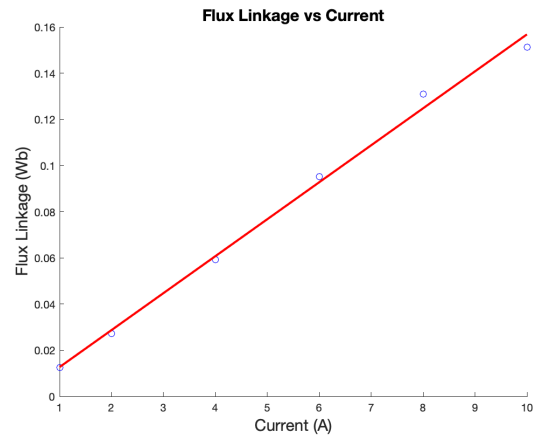


Figure 2: Linear Fit of Flux Linkage vs. Current

C Excitation Parameters Estimation

Armature Resistance (R_f)

For estimating the excitation resistance R_f , I used the same approach as with the armature resistance. More specifically, the resistance was calculated by taking the ratio of the mean voltage to the mean current during the steady-state period. After that, I averaged the estimates across all levels:

$$R_f = \frac{\bar{V}_{\text{mean, steady state}}}{\bar{I}_{\text{mean, steady state}}} = 142.0493 \text{ Ohms}$$

Excitation Inductance (L_f)

To estimate the inductance L_f of the excitation circuit, I initially attempted to model the flux linkage λ_e as a nonlinear function of current, since the relationship between them is expected to be nonlinear in many circuits and is arising from the materials that are used for the excitation circuit. For this purpose, I started by fitting a third-degree polynomial to the flux linkage and current data:

$$p = \text{polyfit}(\text{current levels}, \lambda_e, 3)$$

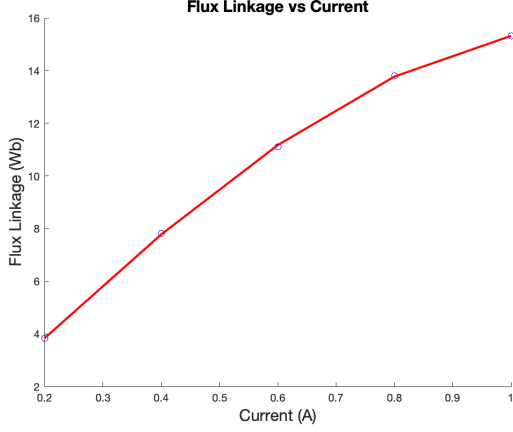


Figure 3: Third-Degree Polynomial Fit of Flux Linkage vs. Current

Moreover, I would like to state that by following this method the general idea was to inverse this function in order to have the output current for every value of flux linkage.

However, I encountered difficulties applying these non-linear characteristics and changes in slope to my Simulink model. Specifically, while attempting to model the non-linear behavior, I realized that the change in slope was only captured between different data sets and not continuously for each current level within the range of 0.2 to 1 A. This meant that my model was not able to dynamically adjust the inductance for every small change in current. So, I assumed that the relationship between the flux linkage and current could be adequately described by a simpler linear model. I proceeded to fit a linear model to the data as presented below:

$$p_{\text{linear}} = \text{polyfit}(\text{current levels}, \lambda_e, 1)$$

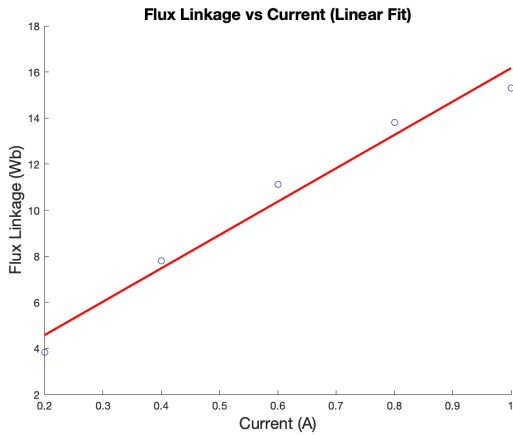


Figure 4: Linear Fit of Flux Linkage vs. Current

Here, like in the armature part, the slope of the

fitted line, represented by $p_{\text{linear}}(1)$, is the estimated inductance L_f of the excitation circuit. It is stated that the relationship between the excitation current and the flux linkage could be described by the formula of: $\lambda = L \cdot I$. The linear fit assumption holds well for the excitation circuit, as the flux linkage increases linearly with the current, which is typical for circuits operating in steady-state conditions under constant excitation. The slope of the line gives the value of the inductance (L_a). Thus,

$$L_f = 14.4854 \text{ H}$$

D $K_f\Phi$

For the calculation of $K_f\Phi$, I processed the recorded measurements, which included different values of voltage and current at a constant speed for various operating points. Using these values and the known armature resistance R_a , I was able to estimate the value of $K_f\Phi$. Specifically, using the formula:

$$K_f\Phi = \frac{u_a - i_a R_a}{\omega_r}$$

This formula is derived from the steady-state equation of a DC motor:

$$u_a = K_f\Phi \cdot \omega_r + i_a R_a$$

I was able to calculate $K_f\Phi$ for each data point and then compute the mean value to get an average estimate.

Thus,

$$K_f\Phi = 1.1128 \text{ Nm}$$

It is worth mentioning that the value of $K_f\Phi$ is very close to the calculated value derived from the nominal parameters of the machine that was aforementioned calculated.

Parameter	Value
Armature Resistance (R_a)	1.5456 Ohms
Armature Inductance (L_a)	16 mH
Excitation Resistance (R_f)	142.0493 Ohms
Excitation Inductance (L_f)	14.4854 H
$K_f\Phi$	1.1128 Nm

Table 1: Summary of Calculated Parameters

III Simulink Model

In this section, the setup of the Simulink model is described, utilizing the estimated parameters. By applying the calculated resistances, inductances, and flux linkage values, the Simulink model aims to replicate the machine's behavior. The model's structure is based on fundamental the principles of

DC machine operation, and MATLAB code is used to define parameter values and simulate it's operating conditions.

The basic equations that describe the operation of a DC machine in the time domain are presented below:

$$u_a = R_a \cdot i_a + L_a \cdot \frac{di_a}{dt} + e_a \quad (1)$$

$$u_f = R_f \cdot i_f + L_f \cdot \frac{di_f}{dt} \quad (2)$$

$$e = K_f \cdot \Phi \cdot \omega \quad (3)$$

$$T_e = K_f \cdot \Phi \cdot i_a \quad (4)$$

$$T_e = J \cdot \frac{d\omega}{dt} + B \cdot \omega + T_{load} \quad (5)$$

In my simulation i want to apply the basic equations in the domain of frequency, so by using the LaPlace transformation I have that:

$$U_a(s) = (R_a + sL_a)I_a(s) + E_a(s) \quad (6)$$

$$U_f(s) = (R_f + sL_f)I_f(s) \quad (7)$$

$$E_a(s) = K_f \Phi \cdot \omega_r(s) \quad (8)$$

$$T_r(s) = K_f \Phi \cdot I_a(s) \quad (9)$$

$$T_r(s) = (B_r + sJ_r)\omega_r(s) + \tau_l \quad (10)$$

In order to describe the three dynamics of the system (armature circuit, excitation circuit, and the mechanical circuit), I applied the mathematical equations using blocks in Simulink. For the numerical values I calculated in the previous section, I want to underline that I assumed a linear relation for the excitation circuit (R-L circuit). In my model, I used all the aforementioned estimated values: R_a , L_a , R_f , and I_f , which were calculated in the previous sections.

Also, I used as inputs the nominal voltage values for the armature ($V_a = 220$ V) and the excitation circuit ($V_f = 145$ V).

Moreover, for my circuit, I defined ϵ . This value expresses the relationship that links the flux linkage with the excitation current. For $K_f \Phi$ and I_f , I used the nominal values:

$$\epsilon = \frac{K_f \Phi}{I_f} = 1.1234 \text{ Nm/A}$$

In the mechanical system, I used a value of $10^{-2} \text{ kg} \cdot \text{m}^2$ for the inertia (J), while for the viscous friction (B), I used a value of $10^{-4} \text{ N} \cdot \text{m} \cdot \text{s/rad}$.

The gain K was used in the mechanical system to transform the output from revolutions per second (rps) to revolutions per minute (rpm):

$$\text{rpm} = \frac{60}{2\pi} \times \text{rps}$$

My DC model system that was created in Simulink is presented in the next figure.

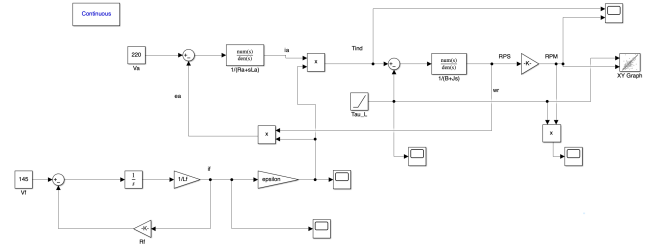


Figure 5: The DC machine model in Simulink

IV Simulation results

This section presents the DC machine's behavior as observed through the Simulink simulation results. The simulated steady-state torque-speed characteristics are compared with theoretical curves derived from the given sets of measurements to evaluate the model's accuracy. This comparison provides insight into the reliability of the estimated parameters and the model's structure and performance, highlighting any discrepancies and discussing potential sources of error.

In the simulation model, I added a constant input to apply a constant torque to the system in order to observe how the speed in RPS of the motor changes in response to the torque. After finding the values from my simulation i tried to find the differences by plotting the torque speed curve in my matlab. In the following sections, I will present my system's results from my simulation and compare them with the results that arise from the theoretical analysis by using matlab.

First of all I would like to mention that i checked 5 values from matlab with 5 values of my simulation.

So, according to my matlab files i took some random values of torque and speed that are presented in the next figure in order to check the results from the simulink model.

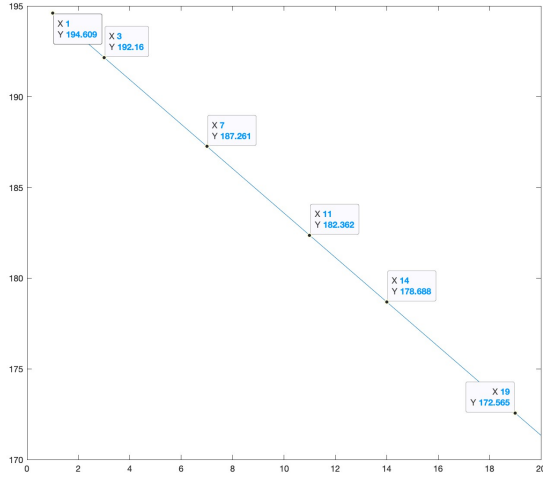


Figure 6: torque-speed curve(y-axis – speed (RPS), x-axis – torque (Nm))

Torque (Nm)	Speed (RPS)
1	194.609
3	192.160
7	187.261
11	182.362
14	178.688
19	172.565

Table 2: Speed vs. Torque

After running my simulation with different constant values of torque, the results that were extracted are presented in the next figure:

Torque (Nm)	Speed (RPS)
1	190.7
3	188.3
7	183.6
11	178.9
14	175.4
19	169.5

Table 3: Simulink results: Speed vs Torque

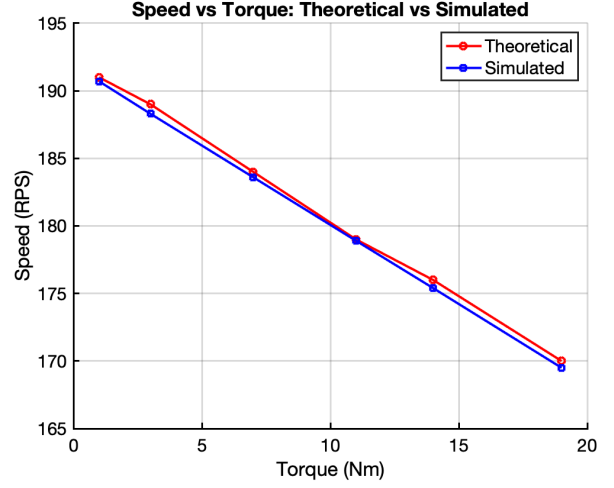


Figure 7: Theoretical and resulting values from the simulations

As we can see the theoretical results from the matlab calculations and the simulations have small differences and the model can describe the operation of the machine.

Another fact that I found interesting and i wanted to investigate for the simulation was to conduct several tests under different conditions to analyze the system's response. In the final figure, I present the results of my simulation using a ramp torque input with a value of 1 Nm. The simulation was set to run for a total time of 100 s.

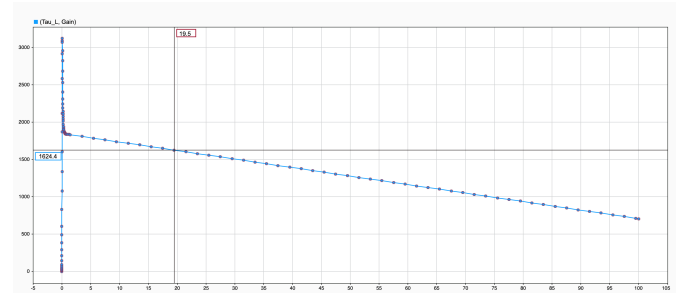


Figure 8: RPM and torque diagram/torque-speed curve(y-axis – RPM (), x-axis – torque ())

We can observe how the machine's speed varies with changes in torque. Notably, there is a sharp increase in speed at the beginning. Since we have constant input values, the output power $P_{out} = \omega_{out} \cdot T_{out}$ remains steady. However, at startup, because the torque is initially zero, there should be a momentary sharp increase in speed. This occurs because, with no opposing torque, the system accelerates rapidly, resulting in a rapid increase in angular velocity until the torque begins to generate and balance the system dynamics.

V Conclusion

We observe that the results are satisfactory, although there are some differences between the simulation and the results obtained from MATLAB. The approach using linear characteristics might introduce some errors compared to the actual operation, as especially in the excitation circuit, in which there is a nonlinear behavior, as previously mentioned. Finally, I would like to state that discussions with fellow students helped me to clarify some points related to the assignment.

VI References

References

- [1] Stephan J. Chapman, *Electric Machinery Fundamentals*, 5th Edition, Chapter 8, New York: McGraw-Hill, 2012.
- [2] Course material from the lectures and lecture slides for the course.